# Small Head Sets for Simple Tasks

Lexical task heads, function-vector heads and attention filters in Llama-3.1-8B-Instruct

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**Abstract**

We look for small sets of attention heads whose mean output, written into a prompt that does not state the task, makes Llama-3.1-8B-Instruct perform it, on 14 simple tasks. We re-implement two published ways of choosing heads, by the words a head decodes (Yang et al., 2026) and by its causal effect (Davidson et al., 2025), read heads with Head-JLens as well as with the logit lens, and add rules that combine what a head decodes with where it attends. All rules are then compared at the same set sizes, 20 to 80 heads, both as one set per task and as one set for all tasks, with heads chosen on half of each task's items and tested on the other half. Head-JLens finds more heads that carry the task than the logit lens, and the hand-written word lists are the weak point of the lexical method. Per task, heads that attend to the instruction's task words and little to the first token match or beat the causal ranking at every size, without any patching during selection. Ranking heads also by their attention to the rest of the instruction and to its question–answer markers, token types that carry task information in in-context learning (Bai et al., 2026), gives the best per-task sets so far, though they were picked among many combinations on the test items. For one set shared by all tasks the causal ranking is best, once a few generic heads that decode the same tokens on every task are removed. Task types rest on different heads: translations on attention-filtered heads, morphology on causally ranked ones. When the task is asked in 30 other wordings, every set loses some accuracy; chosen on all the wordings, the per-task filter sets and the shared causal set without generic heads win it back, and the filter sets come closer to the shared causal set. Readings and attention are enough to find a small head set for each task; one set for all tasks still needs the causal score.

## 1 Introduction

A task representation in a language model can be located by finding the attention heads whose outputs, averaged over prompts that state the task and written into a prompt that does not, make the model perform the task (Todd et al., 2024). Two recent studies choose those heads in different ways. Yang et al. (2026) decode each head's output at the last prompt token with the logit lens and call a head a *lexical task head* when its top tokens contain words describing the task (*opposite*, *capital*). Davidson et al. (2025) rank heads by their causal effect on the answer when the prompt's instruction is replaced by uninformative text, and sum the mean outputs of the top heads into an *instruction function vector*.

Our goal is the smallest set of heads that represents the tasks well, ideally one set for all of them. We reproduce both methods on the same model and tasks, add a second readout, Head-JLens (Merlo, 2026), and build selection rules from what the reproductions show: word lists taken from what heads decode, and filters on where a head attends. We then compare all rules on equal terms. Every rule ranks the heads and gives its top 20, 40, 60 and 80, once per task and once for all tasks, with heads chosen on one half of each task's items and tested on the other.

The paper is organised linearly: the common setup (Sec. 2), the two reproductions (Sec. 3), the design of the comparison (Sec. 4), its results, including the leading rules on 30 other wordings of each task and filters on further parts of the prompt (Sec. 5), and the approaches we would keep (Sec. 6). Only simple tasks are covered.

## 2 Setup

**Model and tasks.** Llama-3.1-8B-Instruct in bfloat16, the main model of Yang et al. (2026) and the only one with released Head-JLens matrices. The 14 simple tasks shared by both source studies, with the datasets of Todd et al. (2024): antonym, synonym, singular–plural, present–past, country–capital, country–currency, landmark–country, park–country, product–company, next-item, prev-item, English–French, English–German, English–Spanish. We group them as word relations, morphology, facts, sequences and translations. Prompts

follow Yang et al. (2026): raw text without the chat template; five instruction templates per task (the cloze form "The antonym of {x} is" and four of the form "<instruction> Q: {x}\nA:"); example prompts with 1 to 30 shots (" $x_1$: $y_1$; x:"); and the zero-shot prompt " x:", which states no task. Decoding is greedy.

**Reading a head.** For head $h$ of layer $l$, $z_{l,h}$ is its output at the last prompt token before the output projection and $c_{l,h} = W_O^{(h)} z_{l,h}$ its write to the residual stream. With $q_v$ the unembedding row of token $v$ scaled by the final norm's gain, the two readouts score $v$ by

$$\text{Logit-head: } q_v^\top c_{l,h}, \qquad \text{J-head: } q_v^\top J_{l,h}\, z_{l,h},$$

where $J_{l,h}$ is the head's part of the Jacobian map of Head-JLens, fitted on WikiText-103 (Merlo, 2026). J-head reads the head through the layers that follow it; Logit-head is the readout of Yang et al. (2026). A head *decodes* a word on a prompt when one of the word's token forms is among its 10 highest-scoring tokens.

**Testing a head set.** A set is tested by mean patching (Yang et al., 2026, Sec. 2.3.2). For each head in the set, its mean output over prompts that state the task (correctly answered, items not patched) is computed; at the last token of the patched prompt, the head's output is replaced by $\alpha$ times this mean, and the model generates its answer. There are two checks.

- **Check 1** patches the zero-shot prompts of items the model answers correctly with one shot. The prompt says nothing about the task, so unpatched accuracy is near zero, and a gain means the set brings the task in. This is the main test, and the only one in the capped comparison.
- **Check 2** patches the 1-shot prompts of items the model answers *wrongly* with one shot (unpatched accuracy 0). The prompt already shows the task and the answer format, so this check measures repair rather than transfer. It is used in the reproductions only.

A *control* patches the same heads with the mean of a related task (the pairing of Yang et al., 2026, e.g. antonym with synonym); a set carries its task when its own mean helps and the control mean does not.

**Scoring.** A reply is correct when it begins with the whole reference answer as words, ignoring case, accents, punctuation and a leading article. This replaces the first-token match of Yang et al. (2026), which marks any number correct on next- and prev-item and a capitalised German noun wrong. The same score selects the correctly answered prompts used everywhere else.

**Held-out protocol.** In the comparison of selection rules (Secs. 4 and 5), every quantity that depends on items is chosen on one half of each task's items and tested on the other. The items are shuffled once (seed 0) and split into halves $A$ and $B$; fold 1 chooses on $A$ and tests on $B$, fold 2 the reverse. "Chosen" covers the prompts whose heads are read, the word lists, the generic heads, the head rankings, the causal scores of Davidson et al., and the task and control means. The means are rule-style: one per instruction template, over its first 100 correctly answered prompts of the choosing half; each patched item is patched with each of the five means and accuracy is averaged over them. Up to 100 items per task and fold are patched (as few as 58 on next-item). The scaling is $\alpha \in \{1, 2\}$. For each fold, $\alpha$ is the value that was better on the other fold for the same task and set (cross-fitting; ties go to $\alpha = 1$), so it is never chosen on the items it is scored on. Random sets are scored at the $\alpha$ of the set they are matched to. Reported numbers are means of the two folds. Replies stop at a newline, the example delimiter or 16 tokens. Every run is single and greedy, and we report accuracies only, without intervals.

# 3 Reproducing the two source methods

## 3.1 Lexical task heads (Yang et al., 2026)

**Method.** A head is a lexical task head for a task and prompt style if, on at least 10% of 100 correctly answered prompts, at least one of its top-10 Logit-head tokens is in a hand-written term list for the task (2 to 22 tokens). The prompts are those of the most accurate instruction template (rule style) or the 5-shot prompts (example style). The set is then tested by mean patching with each wording's own mean, at 15 values of $\alpha$ from $-10$ to 10.

**Fidelity.** We re-run their code's data, templates, term lists, control pairs and parameters (Table 1). We depart from it in four places, each by choice: the padding mask, the J-head readout, the answer score, and bfloat16 instead of float32, the precision in which the lens was fitted. Their code passes batches of 20 left-padded prompts without an attention mask, so the model attends to the padding; we mask it. Run in their

code's own setting (padding attended, first-token score), our Logit-head counts are within one head of their Figs. 79 and 88 on every task (rule style: 250 heads over the 14 tasks against their 247; example style: 127 against 130).

| Setting | Yang et al. (2026) | Here |
|---|---|---|
| Model, precision | Llama-3.1-8B-Instruct, float32 | same model, bfloat16 |
| Tasks, templates, term lists | 14 Todd et al. tasks; first 5 of the 6 templates; their lists | same |
| Batching | 20 left-padded prompts, padding attended | same batches, padding masked |
| Readout | Logit-head | Logit-head and J-head |
| Head criterion | top 10, $\geq$1 term, $\geq$10% of the first 100 correct prompts | same |
| Answer score | first generated token | generated reply, flexible score |
| Means, $\alpha$, control | one mean per wording; 15 values $-10..10$; their control map | same |
| Check 1 | zero-shot prompts (paper, App. G.1); no released script | as the paper |
| Check 2 | 1-shot prompts answered wrongly (the released script) | same |

**Table 1:** Lexical task heads: their setting and ours.

**What reproduces.** Logit-head finds lexical task heads for every task. In check 1, which their paper describes, patching gains only on the fact tasks (and a few more with example means) and does nothing on antonym, synonym, present–past, prev-item and the translations. The per-task curves of their App. G.1 (Figs. 57 and 62), which the text attributes to zero-shot patching, reproduce only in check 2, the setting of their one released causal script. There every task gains, with the largest and smallest gains on the same tasks as in their figures. Their headline finding thus holds in check 2. In check 1, the lexical heads carry only some task types.

**J-head against Logit-head** (Table 2). J-head finds 50 to 70% more heads; most of the added ones lie in layers 10–19, where Logit-head finds few. The heads only J-head finds have causal effects of their own, and J-head sets are at least as effective as Logit-head sets in both checks and both styles. The lead is not a matter of set size: comparing the top 5, 10 and 20 heads of each readout, most of J-head's leads remain and none reverses, while none of Logit-head's few leads remains. J-head does not contain the Logit-head set (only on 4 of 28 task–style pairs). From here on all sets are read with J-head.

| | Heads (sum over tasks) | | | Check 1 effect | | | Check 2 effect | | |
|---|---|---|---|---|---|---|---|---|---|
| Style | L | J | L∩J | L | J | J only | L | J | J only |
| Rule | 240 | 370 | 199 | 0.17 | 0.19 | 0.10 | 0.26 | 0.32 | 0.27 |
| Example | 141 | 237 | 107 | 0.22 | 0.32 | 0.24 | 0.33 | 0.43 | 0.38 |

**Table 2:** Lexical task heads under both readouts (items not held out). Heads: summed over the 14 tasks. Effect: highest accuracy over the 15 values of $\alpha$, averaged over the style's five means, minus unpatched accuracy; mean over tasks. J only: heads J-head finds and Logit-head does not.

## 3.2 Instruction function vectors (Davidson et al., 2025)

**Method.** Each head is scored by its causal indirect effect: the gain in probability of the answer's first token when that head alone is set to its mean output, on prompts whose instruction is replaced by uninformative text of matched length and log-probability. There are three kinds of such text: token sequences sampled to be as probable as the instruction, WikiText-103 prefixes, and other tasks' instructions. The score is averaged over kinds and tasks, and the top 20 heads form one set for all tasks. A task's function vector (FV) is the sum of those heads' mean outputs, added to the residual stream after layer 11 ($|L|/3$) with weight 1.

**Fidelity.** We implement the method from the paper; no public code exists, and the instruction-FV accuracies for this model appear only in a figure, so we have no published per-task numbers to compare with.

Table 3 lists the differences. The main ones are forced by our setting: our instructions are Yang et al.'s templates, and the FV is evaluated in our check 1.

| Setting | Davidson et al. (2025) | Here |
|---|---|---|
| Model, tasks | several models incl. Llama-3.1-8B-Instruct; 50 Todd et al. datasets | Llama-3.1-8B-Instruct; our 14 tasks |
| Instructions | 5 best of LLM-generated instructions, short and long | Yang et al.'s templates 1–4 (template 0 has no separate instruction) |
| Mean outputs | 100 correct prompts, 20 per instruction | 100 correct prompts, 25 per template |
| Uninformative text | 3 kinds, matched in length and log-probability; 25 prompts per kind; 2 instruction lengths | same 3 kinds and matching; 20 prompts per kind; one length; smaller WikiText pool |
| Head score | first-token probability gain, averaged over kinds and tasks | same |
| Heads in the FV | top 20, one set for all tasks | top 30 over the 14 tasks; 20 to 80 in the comparison |
| Intervention | sum of means added after layer 11, weight 1 | same at the last prompt token; also other weights |
| Evaluation | zero-shot, no instruction, their prompt | check 1 (" x:"), flexible score, control mean |

**Table 3:** Instruction function vectors: their setting and ours.

**What we find.** The FV works at their weight: accuracy peaks at $\alpha = 1$ and vanishes from $\alpha = 3$ on. It does nothing on present–past, prev-item and synonym and little on the translations. Reading its 30 heads with J-head shows why the weight is so constrained. Some of them, which we call *generic heads*, decode the same tokens on every task (line ends, colons), attend to the answer slot, and add one large direction shared by all tasks (App. D). The comparison below defines the generic heads on the choosing half alone (Sec. 4) and tests the FV, with and without them, on held-out items.

In the comparison, Davidson et al.'s score ranks heads per task and over the 14 tasks, and its top heads are patched head by head like every other set (*Causal*). Their own intervention, the summed vector, is run on the ranking over the 14 tasks (*Davidson FV*).

# 4 Comparison design

Every rule is compared at the same set sizes, per task and shared, on the same held-out items. A set is defined by its rule, its scope and its size.

- **Size.** Each rule ranks the heads and the set is its top $N$, with $N \in \{20, 40, 60, 80\}$. Ties go to the lower layer, then head.
- **Scope.** *Per task*: heads are ranked on the task's own scores. *Shared* (one set for all 14 tasks): heads are ranked on scores combined over the tasks; each task is still patched with its own means.
- **Rule.** Our decoding and attention filters, Davidson et al.'s causal score as the reference, Yang et al.'s term list as the baseline, and random heads as the control (Table 4). The rules differ in what they need: the lens only, attention patterns (one forward pass per prompt), or a causal score (one patched forward pass per head and prompt, 1,024 heads).

**Names.** Filter sets are named by their letters (D, B, T), the causal sets *Causal*, the baseline *Yang terms*; the size is the number $N$ (DBT-40). *Shared* marks one set for all tasks. $-$ *generic* marks a ranking from which the generic heads were removed before taking the top $N$, so the set still has $N$ heads. *FV* marks Davidson et al.'s own intervention, the heads' means summed into one vector added to the residual stream after layer 11; every other set is patched head by head.

**Task words.** One list per task and fold, used by both D and T, built from the choosing half alone:

1. every head's 10 highest-scoring J-head tokens at the last token of the correctly answered prompts of the five rule templates (up to 100 per template);
2. a token counts as a word if, trimmed and lower-cased, every character is a letter (any script) or a digit, it is not one of 18 stop words (*the, and, for, with, that, this, …*), and it has at least three characters if it contains a Latin letter;
3. a word's *share* is the fraction of (prompt, head) pairs whose top 10 holds it;

| Rule | A head ranks high if … | Needs |
|---|---|---|
| *Decoding and attention filters* | | |
| D | it decodes a word of the task's list on many prompts | lens |
| B | it attends little to the first token (BOS) | attention |
| T | it attends to the task words of the instruction | lens, attention |
| DB, DT, BT, DBT | its ranks on the named filters are good on average | lens, attention |
| DBT − generic | as DBT, the generic heads removed | lens, attention |
| *Davidson et al. (reference)* | | |
| Causal | its causal score is high (patched head by head) | causal |
| Causal − generic | as Causal, the generic heads removed | causal |
| Davidson FV (± generic) | as Shared Causal (± generic); the means summed into one vector after layer 11 (shared only) | causal |
| *Baseline and control* | | |
| Yang terms | it decodes a term of Yang et al.'s hand-written list on many prompts | lens, list |
| Random | drawn at random, as many heads in each layer as DBT of the same scope and $N$ (5 draws) | – |

**Table 4:** The rules of the comparison. Each is run per task and shared, at $N = 20$, 40, 60 and 80 heads.

4. the 50 words with the largest share on 1,000 WikiText-103 passages, read the same way, are *generic words* (*which, what, http, mph, maybe, colour, …*);
5. the task's list is its 30 words with the largest share that are not generic words.

The lists hold the task's own vocabulary (*opposite*, *capital*, *currency*, month names, language names, 法国 on English–French) together with some word pieces (*lingu*, *oppos*) and numbers. App. B gives every list.

**Filter scores.** Over the same prompts, at the last token: **D**, the share of prompts whose top 10 holds a word of the list; **B**, the mean attention on BOS (lower ranks higher); **T**, the mean attention on the task-word positions, the tokens of the instruction (outside the query) that match a word of the list. Three templates contain no list word (the cloze templates of antonym and present–past, and present–past's third instruction); there T is zero for every head, so those tasks are ranked on their other templates. Per task, each filter's ranks are averaged over the combination's filters. Shared sets average ranks over tasks: each filter's within-task ranks are averaged over the 14 tasks, ranked again, and averaged over the combination's filters. Because sets are ranked rather than thresholded, a combination set at large $N$ also holds heads that rank well overall but poorly on one filter.

**Causal score and generic heads.** Davidson et al.'s score (Sec. 3) on the choosing half: per task the task's score, shared the score averaged over tasks. *Generic heads* are defined among the fold's 30 top heads by the shared score. Each head's J-head generic score (the cosine between the 14 tasks' counts of its top-10 tokens, averaged over task pairs; five templates) is sorted, and the 30 scores are cut into a high and a low group with the least spread within each. This marks 9 heads in fold 1 and 8 in fold 2 (L12H9, L13H31, L14H11, L14H29, L15H5, L15H7, L16H29, L31H14, and L28H10 in fold 1); if it marked more than 15, none would be removed. The generic heads are removed from the Causal and DBT rankings before the top $N$ is taken. They are not removed from every head: the same score marks over half of all heads, most of which decode task-independent tokens but are never chosen.

**Baseline and control.** **Yang terms** ranks heads by the share of prompts whose top 10 holds a term of Yang et al.'s hand-written list (their token expansion), on the same prompts; shared, by the within-task ranks averaged over tasks. Heads that never decode a term are not used, so on seven tasks the 60- and 80-head sets hold fewer heads (as few as 45); these are compared with the other rules at the size nearest to theirs. **Random** draws as many heads in each layer as DBT of the same scope and $N$, outside it, five times (shared: one draw for all tasks), and is scored at DBT's $\alpha$.

**Protocol.** Check 1 only, on the held-out items of Sec. 2, $\alpha \in \{1, 2\}$ cross-fitted between the folds, and the control mean for every non-random set: 128 head-patched sets per task and fold, and Davidson et al.'s summed vector at the four sizes with and without the generic heads. Rerunning a set can change a few greedy answers when its batches are made up differently. The per-task Causal-20 and -40 sets were first run with ten values of $\alpha$ per batch; here, with two, up to three prompts of 100 change their answer on some templates,

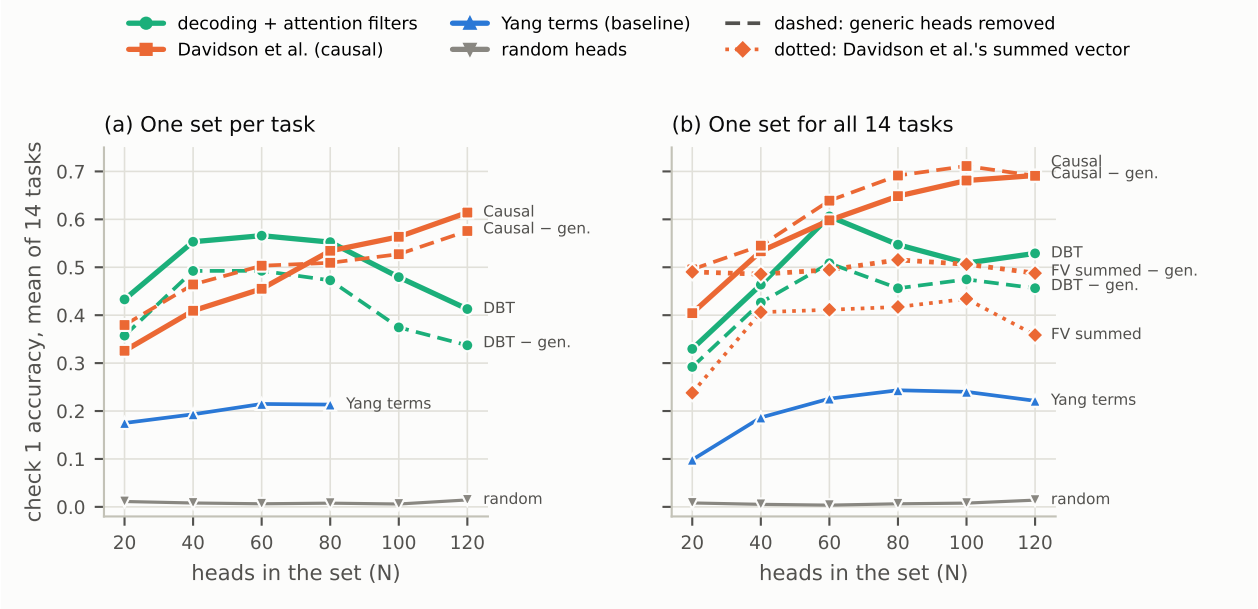



**Figure 1:** Check 1, mean accuracy over the 14 tasks against the number of heads, held-out items, cross-fitted $\alpha \in \{1, 2\}$, mean of the two folds. (a) One set per task; (b) one set for all 14 tasks. Solid: the ranking as it is; dashed: the generic heads removed before taking the top $N$; dotted: Davidson et al.'s summed vector (shared only). Random: layer-matched to DBT, five draws. The other filter combinations are in Table 5.

which moves the 14-task mean by at most 0.004. The shared Causal-20 and -40, first run with the same two values, give identical answers.

# 5 Results

## 5.1 Accuracy against set size

Figure 1 and Table 5 give each rule's mean over the 14 tasks at each size; App. A gives every task at 40 heads and the control means.

- **Per task, the filters match or beat the causal ranking at every size.** DBT and BT lie above Causal, with or without the generic heads, from 20 to 60 heads and level with it at 80. They reach their level at 40 heads and gain little after, whereas the causal ranking keeps rising with size.
- **For one set shared by all tasks, the causal ranking is best.** Shared Causal − generic is the best set at every size and still rising at 80 heads, the highest accuracy of the comparison. Shared DBT and BT reach the shared causal ranking with its generic heads at 60 heads, stay below the one without them, and fall back at 80.
- **The two scopes favour different rules.** The causal score does better as one shared ranking than as per-task rankings at every size. DBT and BT do better per task at 20 and 40 heads, and about as well in both scopes at 60 and 80. At 40 heads, the per-task DBT sets are level with Shared Causal − generic.
- **The hand-written list stays the weak point.** Yang terms is below every other rule except B alone, at every size and in both scopes, and per task it gains little from more heads.
- **Which heads matters, not how many.** Random heads with the same number of heads per layer as DBT do nothing at any size. Control means stay low for every set (Table 10), so the gains are task-specific.

## 5.2 Which filters matter

- **T is the strongest single filter.** Attention to the instruction's task words alone ranks heads better than decoding the task's words (D) or avoiding BOS (B), per task at every size and shared from 40 heads on. Per task, B alone is the weakest filter. D alone carries part of the task, and its shared ranking stops improving at 40 heads.
- **B and T together give most of the filters' effect.** BT and DBT are nearly equal in both scopes: D adds nothing once B and T are used. T's task words still come from the lens, so the lens is needed, though not as a filter. DT, without B, is below both. DB is level with BT and DBT or slightly above at 20 heads (shared also at 40), and below them at larger sizes.

| | One set per task | | | | | | One set for all 14 tasks | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Heads (N) | 20 | 40 | 60 | 80 | 100 | 120 | 20 | 40 | 60 | 80 | 100 | 120 |
| *Decoding and attention filters* | | | | | | | | | | | | |
| D | 0.27 | 0.37 | 0.41 | 0.43 | 0.40 | 0.39 | 0.24 | 0.29 | 0.29 | 0.29 | 0.22 | 0.20 |
| B | 0.13 | 0.22 | 0.23 | 0.23 | 0.22 | 0.23 | 0.11 | 0.25 | 0.41 | 0.40 | 0.44 | 0.45 |
| T | 0.31 | 0.45 | 0.48 | 0.54 | **0.57** | 0.58 | 0.23 | 0.43 | 0.44 | 0.50 | 0.56 | 0.53 |
| DB | **0.44** | 0.45 | 0.36 | 0.32 | 0.27 | 0.24 | 0.37 | 0.47 | 0.54 | 0.52 | 0.45 | 0.46 |
| DT | 0.35 | 0.47 | 0.49 | 0.51 | 0.53 | 0.54 | 0.32 | 0.37 | 0.45 | 0.48 | 0.46 | 0.42 |
| BT | 0.42 | 0.54 | **0.58** | 0.55 | 0.57 | 0.57 | 0.35 | 0.45 | 0.60 | 0.56 | 0.58 | 0.63 |
| DBT | 0.43 | **0.55** | 0.57 | **0.55** | 0.48 | 0.41 | 0.33 | 0.46 | 0.61 | 0.55 | 0.51 | 0.53 |
| DBT − generic | 0.36 | 0.49 | 0.49 | 0.47 | 0.37 | 0.34 | 0.29 | 0.43 | 0.51 | 0.46 | 0.47 | 0.46 |
| *Davidson et al. (causal score)* | | | | | | | | | | | | |
| Causal | 0.33 | 0.41 | 0.46 | 0.53 | 0.56 | **0.61** | 0.40 | 0.53 | 0.60 | 0.65 | 0.68 | **0.69** |
| Causal − generic | 0.38 | 0.46 | 0.50 | 0.51 | 0.53 | 0.58 | **0.50** | **0.55** | **0.64** | **0.69** | **0.71** | 0.69 |
| Davidson FV (summed) | – | – | – | – | – | – | 0.24 | 0.41 | 0.41 | 0.42 | 0.43 | 0.36 |
| Davidson FV − generic (summed) | – | – | – | – | – | – | 0.49 | 0.49 | 0.49 | 0.52 | 0.51 | 0.49 |
| *Baseline* | | | | | | | | | | | | |
| Yang terms | 0.18 | 0.19 | 0.21 | 0.21 | – | – | 0.10 | 0.19 | 0.23 | 0.24 | 0.24 | 0.22 |
| *Control* | | | | | | | | | | | | |
| Random (DBT layers) | 0.01 | 0.01 | 0.01 | 0.01 | 0.01 | 0.01 | 0.01 | 0.01 | 0.00 | 0.01 | 0.01 | 0.01 |

**Table 5:** Check 1, mean accuracy over the 14 tasks, held-out items, cross-fitted $\alpha \in \{1, 2\}$, mean of the two folds. Bold: the best non-random set in each column. Davidson FV: their summed vector, run on the shared ranking only. Yang terms holds fewer heads than $N$ on some tasks at 60 and 80 (58 and 72 heads on average over tasks and folds). Random: layer-matched to DBT, five draws, at DBT's $\alpha$. Unpatched accuracy: 0.01.

- **Decoding without T fails at larger sizes.** Per task, DB stays level from 20 to 40 heads and then falls, below D alone at 60 and 80. The loss is mostly on the fact tasks: with the larger DB sets the model describes the query ("Located in the heart of …", "A UNESCO World Heritage …" on park–country) instead of naming the answer. DBT sets of the same size, which add attention to the task words, give the answer. Shared DB does not fall.
- **The filters and the causal score partly agree.** Per task, DBT shares about a third of its heads with Causal of the same size; shared, about half. The median layer of both is about 14; that of the sets that only decode (D, Yang terms) is 16 to 18.

## 5.3 Generic heads and Davidson et al.'s summed vector

- **Removing the generic heads helps the causal rankings**, in both scopes at nearly every size, most for the shared ranking at 20 heads. That set holds five of the generic heads, and from 40 heads on it holds all of them. Removing them also steadies it across folds: with the generic heads the two folds' Shared Causal-20 differ widely, without them they agree.
- **Removing them lowers DBT** at every size and in both scopes. DBT sets hold fewer generic heads (two to five) than the causal sets, and removing these costs accuracy instead of adding it. Whether a generic head harms a set thus depends on the set.
- **Davidson et al.'s summed vector is below head-by-head patching of the same heads.** With the generic heads it is below at every size. From 40 heads on, the cross-fit always picks $\alpha = 1$, because at $\alpha = 2$ the vector fails. At 20 heads it is unstable across folds: at $\alpha = 2$ it is the second-best set of fold 1, behind one with four times its heads, and fails in fold 2, so the cross-fitted mean is low. The two folds' top 20 differ by two heads; fold 2's holds the late generic head L31H14. Without the generic heads, the summed vector is level with head-by-head patching at 20 heads but does not improve with more heads, so head-by-head patching overtakes it from 40 on.

## 5.4 By task type

The means hide large differences between task types (Table 6; per task in Table 9).

- **Facts** (capitals, currency, landmark, park, product) are carried by almost every rule from 20 heads on, shared or not; Yang terms and per-task DB above 40 heads are the exceptions.
- **Morphology** is carried best by the causal rankings, most of all the shared one, which from 60 heads gets nearly every item. Per task, DB and DBT also carry it. T alone does little here below 80 heads:

| | One set per task | | | | | | | One set for all tasks | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Task type | DBT | DB | T | DBT −gen. | Causal | Causal −gen. | Yang | DBT | DBT −gen. | Causal | Causal −gen. | FV sum | FV sum −gen. | Yang |
| *20 heads* | | | | | | | | | | | | | | |
| Word relations (2) | 0.24 | 0.25 | 0.26 | 0.18 | 0.10 | 0.30 | 0.05 | 0.07 | 0.05 | 0.17 | 0.33 | 0.09 | **0.42** | 0.00 |
| Morphology (2) | 0.48 | 0.63 | 0.24 | 0.38 | 0.54 | 0.61 | 0.01 | 0.43 | 0.38 | 0.66 | **0.74** | 0.25 | 0.55 | 0.00 |
| Facts (5) | 0.71 | 0.74 | 0.58 | 0.68 | 0.65 | 0.70 | 0.46 | 0.70 | 0.63 | 0.72 | 0.81 | 0.49 | **0.87** | 0.27 |
| Sequences (2) | 0.04 | 0.00 | 0.02 | 0.03 | 0.01 | 0.00 | 0.01 | 0.00 | 0.01 | 0.06 | **0.09** | 0.02 | 0.01 | 0.00 |
| Translations (3) | **0.33** | 0.22 | 0.13 | 0.14 | 0.00 | 0.00 | 0.00 | 0.03 | 0.01 | 0.09 | 0.19 | 0.05 | 0.18 | 0.00 |
| *40 heads* | | | | | | | | | | | | | | |
| Word relations (2) | 0.36 | 0.45 | 0.30 | 0.36 | 0.26 | **0.47** | 0.08 | 0.18 | 0.12 | 0.24 | 0.35 | 0.17 | 0.26 | 0.00 |
| Morphology (2) | 0.64 | 0.70 | 0.27 | 0.56 | 0.65 | 0.78 | 0.01 | 0.53 | 0.57 | 0.73 | **0.81** | 0.36 | 0.62 | 0.02 |
| Facts (5) | 0.80 | 0.56 | 0.74 | 0.75 | 0.73 | 0.75 | 0.47 | **0.87** | 0.82 | 0.79 | 0.84 | 0.75 | 0.86 | 0.50 |
| Sequences (2) | **0.19** | 0.01 | 0.06 | 0.16 | 0.09 | 0.09 | 0.05 | 0.16 | 0.13 | 0.12 | 0.17 | 0.14 | 0.04 | 0.04 |
| Translations (3) | **0.45** | 0.40 | 0.42 | 0.32 | 0.02 | 0.01 | 0.02 | 0.14 | 0.07 | 0.44 | 0.25 | 0.19 | 0.23 | 0.00 |
| *60 heads* | | | | | | | | | | | | | | |
| Word relations (2) | 0.40 | 0.42 | 0.40 | 0.38 | 0.35 | **0.46** | 0.04 | 0.28 | 0.15 | 0.32 | 0.45 | 0.16 | 0.29 | 0.00 |
| Morphology (2) | 0.72 | 0.57 | 0.37 | 0.56 | 0.58 | 0.76 | 0.04 | 0.74 | 0.57 | 0.83 | **0.88** | 0.34 | 0.58 | 0.00 |
| Facts (5) | 0.82 | 0.39 | 0.79 | 0.76 | 0.78 | 0.79 | 0.51 | **0.89** | 0.85 | 0.82 | 0.86 | 0.74 | 0.84 | 0.61 |
| Sequences (2) | 0.11 | 0.03 | 0.07 | 0.10 | 0.19 | 0.22 | 0.11 | 0.26 | 0.21 | 0.17 | **0.29** | 0.16 | 0.00 | 0.04 |
| Translations (3) | 0.46 | 0.32 | 0.38 | 0.33 | 0.08 | 0.07 | 0.01 | 0.49 | 0.34 | **0.54** | 0.47 | 0.25 | 0.32 | 0.00 |
| *80 heads* | | | | | | | | | | | | | | |
| Word relations (2) | 0.40 | 0.44 | 0.43 | 0.34 | 0.44 | 0.46 | 0.01 | 0.23 | 0.14 | 0.45 | **0.55** | 0.17 | 0.34 | 0.00 |
| Morphology (2) | 0.72 | 0.40 | 0.63 | 0.61 | 0.73 | 0.64 | 0.04 | 0.66 | 0.47 | 0.90 | **0.92** | 0.38 | 0.55 | 0.00 |
| Facts (5) | 0.78 | 0.41 | 0.80 | 0.70 | 0.81 | 0.81 | 0.48 | 0.84 | 0.79 | 0.83 | **0.86** | 0.72 | 0.84 | 0.66 |
| Sequences (2) | 0.12 | 0.04 | 0.11 | 0.10 | 0.35 | 0.28 | 0.20 | 0.31 | 0.26 | 0.23 | **0.36** | 0.21 | 0.07 | 0.04 |
| Translations (3) | 0.45 | 0.22 | 0.43 | 0.34 | 0.13 | 0.10 | 0.02 | 0.36 | 0.23 | **0.58** | 0.58 | 0.23 | 0.36 | 0.00 |
| *100 heads* | | | | | | | | | | | | | | |
| Word relations (2) | 0.41 | 0.34 | 0.41 | 0.33 | 0.48 | 0.45 | – | 0.22 | 0.18 | 0.53 | **0.59** | 0.17 | 0.32 | 0.01 |
| Morphology (2) | 0.60 | 0.33 | 0.64 | 0.46 | 0.83 | 0.71 | – | 0.58 | 0.48 | **0.95** | 0.88 | 0.47 | 0.56 | 0.01 |
| Facts (5) | 0.69 | 0.38 | 0.81 | 0.57 | 0.82 | 0.81 | – | 0.81 | 0.78 | 0.86 | **0.86** | 0.70 | 0.84 | 0.65 |
| Sequences (2) | 0.07 | 0.04 | 0.12 | 0.05 | 0.38 | 0.31 | – | 0.25 | 0.25 | 0.20 | **0.41** | 0.22 | 0.07 | 0.05 |
| Translations (3) | 0.37 | 0.15 | 0.52 | 0.24 | 0.14 | 0.13 | – | 0.33 | 0.31 | **0.64** | 0.63 | 0.27 | 0.32 | 0.00 |
| *120 heads* | | | | | | | | | | | | | | |
| Word relations (2) | 0.38 | 0.24 | 0.42 | 0.31 | **0.53** | 0.47 | – | 0.25 | 0.18 | 0.51 | 0.51 | 0.11 | 0.22 | 0.00 |
| Morphology (2) | 0.45 | 0.36 | 0.64 | 0.40 | 0.89 | 0.85 | – | 0.61 | 0.46 | **0.95** | 0.83 | 0.40 | 0.57 | 0.01 |
| Facts (5) | 0.61 | 0.33 | 0.83 | 0.50 | 0.82 | 0.81 | – | 0.79 | 0.75 | **0.86** | 0.85 | 0.64 | 0.84 | 0.58 |
| Sequences (2) | 0.07 | 0.03 | 0.10 | 0.06 | 0.43 | 0.35 | – | 0.20 | 0.21 | 0.23 | **0.44** | 0.11 | 0.06 | 0.04 |
| Translations (3) | 0.31 | 0.14 | 0.58 | 0.23 | 0.26 | 0.21 | – | 0.43 | 0.32 | **0.66** | 0.64 | 0.20 | 0.32 | 0.02 |

**Table 6:** Check 1, mean accuracy per task type (number of tasks in brackets) at each set size; held-out items, cross-fitted $\alpha \in \{1, 2\}$, mean of the two folds. Bold: highest in the row. BT, close to DBT, is in Table 5 only.

present–past has no task word in two of its five templates.

- **Translations** are carried by the per-task sets that use attention (BT and DBT at every size), and by no per-task causal ranking at any size. Ranked over all tasks, the causal score does carry them, from 40 heads on, and so do the shared BT and DBT sets from 60. Removing the generic heads lowers the shared causal sets on translations at 40 and 60 heads.
- **Sequences** (next-, prev-item) stay low under every rule. They gain most at 60 to 80 heads, with the causal rankings (per task and shared) and Shared DBT.
- **Word relations** are carried best by Causal − generic in both scopes from 40 heads on, by the summed vector without generic heads at 20, and per task also by DB.

No rule wins every type. Per task, the filters win translations, and the causal ranking without generic heads wins word relations and, up to 60 heads, morphology. No rule carries sequences.

### 5.5 Other wordings

Every set above was chosen and tested on Yang et al.'s five templates. To test the sets on the task asked in other words, each task gets 30 more templates, generated by GPT-5.5 and checked by script, not by hand. They come in six groups of five: three forms (cloze, "An antonym for {x} would be"; question–answer, "Provide the antonym for the given word. Q: {x}\nA:"; label, "Country: {x}\nCapital city:") × whether the template *names* the task, that is, holds a term of Yang et al.'s list or the task's name ("The contrasting word for {x} is" does not). Without a patch, the model answers the question–answer and label templates that name the task about as well as Yang et al.'s, and the cloze templates and those that do not name the task less well (Table 12).

**Design.** DBT, Causal and Causal − generic, per task and shared, at 40 and 80 heads, in two versions. *Chosen on Yang's 5*: the sets above, unchanged. *Chosen on all 35*: the same rules with every quantity of the choosing half computed on the 35 templates: the word lists, D, B and T from their correctly answered prompts (up to 100 per template); Davidson et al.'s causal score on the 14 templates of the question–answer form (the cloze and label templates have no instruction to replace); and the generic heads by the two-group split of the generic score read on all 35. Every set is patched with each template's own mean, one at a time, and accuracy is averaged over the templates; a template with no correctly answered prompt on the choosing half gives no mean there (20 of the 420). $\alpha$ is cross-fitted on this average, as above. Random sets are layer-matched to DBT-80 chosen on all 35. Check: the sets chosen on Yang's five, patched with Yang's five means only, give the numbers of Table 5 exactly.

- **The sets lose accuracy on the new wordings.** Every set chosen on Yang's five is lower with the means of all 35 templates than with Yang's five (Table 7), most of all with cloze templates that do not name the task (Table 13). The order of the rules does not change.
- **Choosing on all 35 templates wins the loss back, except for the causal ranking with its generic heads.** DBT and Causal − generic chosen on all 35 lie above the same rules chosen on Yang's five, in both scopes and at both sizes, and on most tasks (at 40 heads, per-task DBT and shared DBT and Causal − generic on every task). They are higher on Yang's five templates as well, so more templates make the choice better, not only closer to the new wordings. Causal with its generic heads gains a little per task and loses a little shared.
- **Per task, DBT is the best rule; shared, Causal − generic.** Chosen on all 35, per-task DBT is level at 40 and 80 heads, above the per-task causal rankings at both sizes, and level with Shared Causal − generic-40. Shared Causal − generic-80 stays the best set, with a smaller lead over the DBT sets than on Yang's five. Shared DBT gains most at 40 heads, which brings it close to 80.
- **Removing the generic heads matters more.** On all 35 templates it raises every causal ranking, most of all the shared one at 40 heads. The generic heads marked on all 35 are mostly those marked on Yang's five (App. C).
- **All sets follow the form of the wording.** Every set does best with the means of Yang's five, of the question–answer templates that name the task and, for DBT, of the label templates that name it, and worst with cloze templates that do not name it, in about the order of the model's own accuracy on them (Table 12). The spread is widest for the causal rankings with their generic heads, which on cloze templates keep about half their question–answer accuracy or less; DBT and Causal − generic are more even.
- **By task** (Table 14): English–Spanish and antonym gain most from choosing on all 35 templates; the per-task causal rankings now carry part of English–Spanish, but little of French and German. The fact tasks other than country–currency, already high, barely change. Sequences stay low, and per-task Causal on next-item falls to almost nothing.
- **Re-choosing changes the causal sets most** (Table 16): about three quarters of the DBT heads stay, a little over half of the per-task causal heads, and about four fifths of the shared causal heads. Control

| | One set per task | | | One set for all 14 tasks | | |
|---|---|---|---|---|---|---|
| Heads chosen on | Yang's 5 | Yang's 5 | all 35 | Yang's 5 | Yang's 5 | all 35 |
| Means from | Yang's 5 | all 35 | all 35 | Yang's 5 | all 35 | all 35 |
| *40 heads* | | | | | | |
| DBT | 0.55 | 0.50 | **0.57** | 0.46 | 0.43 | **0.55** |
| Causal | **0.41** | 0.37 | 0.40 | **0.53** | 0.46 | 0.45 |
| Causal − generic | 0.46 | 0.40 | **0.49** | 0.55 | 0.47 | **0.57** |
| *80 heads* | | | | | | |
| DBT | 0.55 | 0.49 | **0.57** | 0.55 | 0.49 | **0.57** |
| Causal | **0.53** | 0.44 | 0.48 | **0.65** | 0.56 | 0.54 |
| Causal − generic | 0.51 | 0.45 | **0.52** | **0.69** | 0.61 | 0.63 |
| Random (DBT-80 layers) | – | – | 0.01 | – | – | 0.00 |
| *100 heads* | | | | | | |
| DBT | 0.48 | 0.43 | **0.57** | 0.51 | 0.46 | **0.56** |
| Causal | **0.56** | 0.47 | 0.52 | **0.68** | 0.59 | 0.57 |
| Causal − generic | 0.53 | 0.44 | **0.54** | **0.71** | 0.62 | 0.65 |
| *120 heads* | | | | | | |
| DBT | 0.41 | 0.37 | **0.55** | 0.53 | 0.46 | **0.59** |
| Causal | **0.61** | 0.51 | 0.53 | **0.69** | 0.60 | 0.60 |
| Causal − generic | **0.58** | 0.48 | 0.55 | **0.69** | 0.58 | 0.61 |
| Random (DBT-120 layers) | – | – | 0.02 | – | – | 0.01 |
| Control mean, largest over these sets | | | 0.04 | | | |

**Table 7:** Check 1 on other wordings, mean accuracy over the 14 tasks, held-out items, cross-fitted $\alpha \in \{1, 2\}$, mean of the two folds. First column at each size: the sets and numbers of Table 5. Means from all 35: each set patched with each template's own mean, accuracy averaged over the templates. Bold: the best of the three versions of a set. Random: layer-matched to DBT-80 chosen on all 35, five draws.

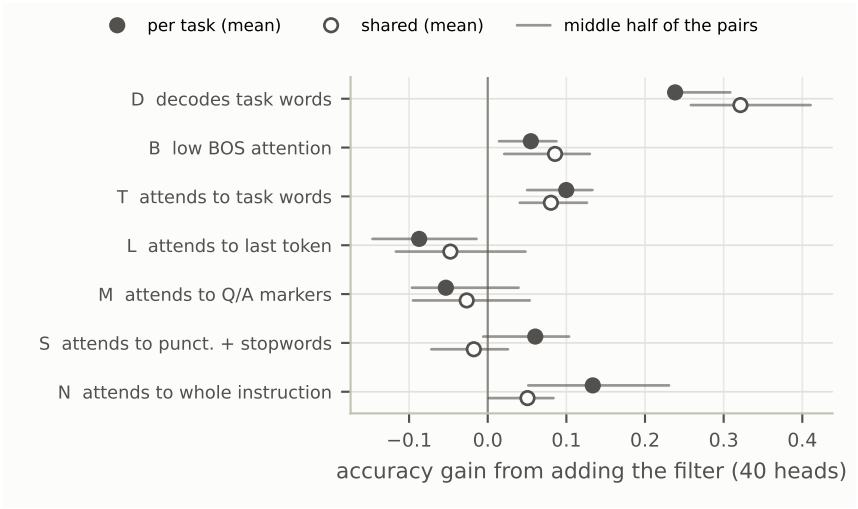



**Figure 2:** What each filter adds at 40 heads: accuracy of a combination with the filter minus that of the same combination without it, over every such pair among the 79 combinations (check 1, held-out items, cross-fitted $\alpha$, mean of the folds and of the 14 tasks). Dots: mean over the pairs; lines: their middle half. A filter alone against no filter is left out. N has fewer pairs than the others, as it never joins T or S.

means stay at or below 0.03 and random sets do nothing.

### 5.6 Attention to other parts of the prompt

T counts only the attention a head pays to the instruction's task words. Bai et al. (2026) find that in in-context learning the model gathers what it needs at the template tokens, punctuation and stopwords of the prompt more than at its content words. We split the rest of the prompt the same way and use the attention to each part as a filter.

**Filters.** Over the prompts and token of the other filters (the last token of the correctly answered prompts of the five rule templates, choosing half), every prompt token is put in one category, in this order: BOS, task words (T's positions), the query, the last token itself, the template markers (" Q:" before the query and "\nA" after it; the cloze templates have none), punctuation, stopwords (NLTK's English list) and the other instruction words. Four new filters, each ranking heads by a head's attention to one part: **L**, the last token itself; **M**, the template markers; **S**, punctuation and stopwords (Bai et al.'s stopword tokens); **N**, the whole instruction (task words, punctuation, stopwords and other words), that is, T without the word list. Unlike Bai et al., who assign categories by hand and mask tokens, we assign them by script and rank heads.

**Design.** Two steps, under the protocol of Sec. 4 (check 1, held-out items, $\alpha \in \{1, 2\}$ cross-fitted, control mean), with ranks combined as there. First, every combination of the seven filters D, B, T, L, M, S and N at 40 heads, per task and shared, except those that hold N together with T or S, whose positions N already counts: 79 combinations per scope. Second, the 15 best combinations of each scope at all six sizes. The 15 are chosen on the held-out accuracy they are then reported on, and each set's best size is read off the same items, so their accuracy is optimistic; the sets of Sec. 5 were fixed before their runs. The random sets of Figure 1 remain the control.

- **Decoding matters most** (Figure 2). Adding D raises nearly every combination, in both scopes, and by more than any attention filter. Among the attention filters, attention to the whole instruction adds most per task, then attention to the task words; low attention to BOS and attention to punctuation and stopwords add less, the latter per task only. Attention to the last token or the template markers lowers accuracy on average, and alone carries almost nothing.
- **Marker and last-token attention help only next to decoding.** Without D, adding M or L lowers most combinations. With D and without S, adding M raises the set in both scopes, and adding L raises the shared sets. The best per-task combination at 40 heads, DBMN, ranks attention without a word list: to the whole instruction and to the markers.
- **Per task, the new combinations are the best sets from 40 heads on** (Figure 3, Table 8). Eleven of the 15 peak above the best set of Table 5, the per-task causal ranking at 120 heads, and DBMN passes it with a third as many heads. With more heads they lose less than DBT, which falls back after 60: DBMN

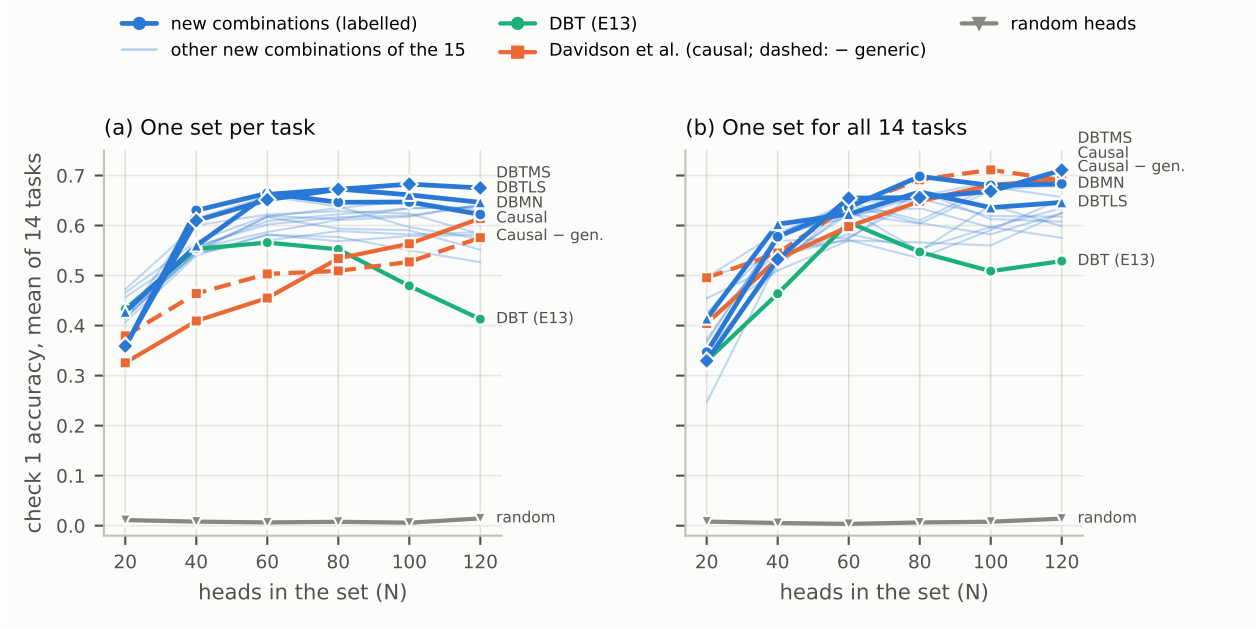



**Figure 3:** Check 1, mean accuracy over the 14 tasks against the number of heads for the 15 best combinations of each scope at 40 heads (blue), held-out items, cross-fitted $\alpha \in \{1, 2\}$, mean of the two folds; DBT, Causal, Causal − generic and random as in Figure 1. Labelled: DBMN (best per task at 40 heads), DBTMS (best at its peak in both scopes) and DBTLS (best shared at 40 heads); the other twelve of each scope in pale blue. The 15 were chosen on these items. Every value is in Table 8.

| **per-task sets (one set per task)** | | | | | | |
|---|---|---|---|---|---|---|
| Set | 20 | 40 | 60 | 80 | 100 | 120 |
| 1. DBMN | 0.36 | **0.63** | **0.66** | 0.65 | 0.65 | 0.62 |
| 2. DBTM | 0.41 | 0.61 | **0.66** | 0.64 | 0.60 | 0.57 |
| 3. DBTMS | 0.36 | 0.61 | 0.65 | **0.67** | **0.68** | **0.68** |
| 4. DBS | **0.47** | 0.60 | 0.62 | 0.63 | 0.62 | 0.58 |
| 5. DBTS | **0.47** | 0.57 | 0.62 | 0.61 | 0.64 | 0.64 |
| 6. DBTLS | 0.43 | 0.56 | **0.66** | **0.67** | 0.66 | 0.65 |
| 7. BTS | 0.43 | 0.56 | 0.60 | 0.62 | 0.62 | 0.64 |
| 8. DS | 0.41 | 0.56 | 0.61 | 0.62 | 0.63 | 0.64 |
| 9. DTS | 0.43 | 0.55 | 0.59 | 0.61 | 0.62 | 0.64 |
| 10. DBT | 0.43 | 0.55 | 0.57 | 0.55 | 0.48 | 0.41 |
| 11. DBLN | 0.46 | 0.55 | 0.62 | 0.59 | 0.59 | 0.55 |
| 12. DBLS | 0.41 | 0.55 | 0.62 | 0.63 | 0.65 | 0.63 |
| 13. BN | 0.42 | 0.55 | 0.57 | 0.59 | 0.58 | 0.59 |
| 14. DBN | 0.43 | 0.54 | 0.58 | 0.58 | 0.55 | 0.53 |
| 15. DLN | 0.41 | 0.54 | 0.58 | 0.57 | 0.58 | 0.58 |
| Causal | 0.33 | 0.41 | 0.46 | 0.53 | 0.56 | 0.61 |
| Causal − generic | 0.38 | 0.46 | 0.50 | 0.51 | 0.53 | 0.58 |
| T (E13) | 0.31 | 0.45 | 0.48 | 0.54 | 0.57 | 0.58 |
| BT (E13) | 0.42 | 0.54 | 0.58 | 0.55 | 0.57 | 0.57 |
| random | 0.01 | 0.01 | 0.01 | 0.01 | 0.01 | 0.01 |

| **shared sets (one set for all 14 tasks)** | | | | | | |
|---|---|---|---|---|---|---|
| Set | 20 | 40 | 60 | 80 | 100 | 120 |
| 1. DBTLS | 0.41 | **0.60** | 0.62 | 0.67 | 0.64 | 0.65 |
| 2. DBTLM | **0.50** | 0.59 | 0.64 | 0.60 | 0.58 | 0.62 |
| 3. DBTM | 0.37 | 0.58 | 0.62 | 0.66 | 0.69 | 0.68 |
| 4. DBTLMS | 0.37 | 0.58 | **0.65** | 0.65 | 0.67 | 0.68 |
| 5. DBMN | 0.35 | 0.58 | 0.64 | **0.70** | 0.68 | 0.68 |
| 6. DBMS | 0.34 | 0.58 | 0.62 | 0.60 | 0.67 | 0.69 |
| 7. DBLMN | 0.43 | 0.57 | 0.58 | 0.65 | 0.62 | 0.62 |
| 8. DLMN | 0.40 | 0.55 | 0.57 | 0.53 | 0.59 | 0.65 |
| 9. DBLMS | 0.40 | 0.54 | 0.57 | 0.67 | 0.61 | 0.61 |
| 10. DBTL | **0.50** | 0.54 | 0.58 | 0.55 | 0.60 | 0.58 |
| 11. DBTMS | 0.33 | 0.53 | **0.65** | 0.66 | 0.67 | **0.71** |
| 12. DBM | 0.25 | 0.53 | 0.62 | 0.63 | 0.60 | 0.63 |
| 13. DBLN | 0.45 | 0.52 | 0.63 | 0.55 | 0.64 | 0.60 |
| 14. DTLMS | 0.35 | 0.52 | **0.65** | 0.61 | 0.68 | 0.66 |
| 15. DTLM | 0.42 | 0.51 | 0.57 | 0.57 | 0.56 | 0.63 |
| Causal | 0.40 | 0.53 | 0.60 | 0.65 | 0.68 | 0.69 |
| Causal − generic | **0.50** | 0.55 | 0.64 | 0.69 | **0.71** | 0.69 |
| T (E13) | 0.23 | 0.43 | 0.44 | 0.50 | 0.56 | 0.53 |
| BT (E13) | 0.35 | 0.45 | 0.60 | 0.56 | 0.58 | 0.63 |
| random | 0.01 | 0.01 | 0.00 | 0.01 | 0.01 | 0.01 |

**Table 8:** Check 1, mean accuracy over the 14 tasks of the 15 best combinations of each scope, ranked by their accuracy at 40 heads, at every size; held-out items, cross-fitted $\alpha \in \{1, 2\}$, mean of the two folds. Rows 1–15 were chosen on these items. Below: sets of Table 5. Bold: best of the column at two decimals. Letters: D decodes task words; B low BOS attention; attention to T task words, L the last token, M the template markers, S punctuation and stopwords, N the whole instruction. Random: layer-matched to DBT.

and DBTM peak at 60, and DBTS, BTS, DS and DTS still rise at 120. At 20 heads they are about level with DBT and BT.

- **Shared, they match the causal ranking without generic heads but do not beat it.** The best new combinations come within about a point of its peak at 80 to 120 heads, lead it at 40 and trail it slightly at 100. The shared curves differ more between the folds than the per-task ones.
- **By task type the new sets follow DBT**: per task they carry translations, facts and most word relations, and stay below the causal rankings on morphology and sequences. Control means stay at or below 0.05.

# 6 Best approaches

For the stated goal, a small set of heads that represents the tasks in general:

1. **Read heads with J-head.** It finds more heads that carry the task, in middle layers the logit lens misses, and its lead holds at equal set size.
2. **One set for all tasks: Shared Causal − generic**, patched head by head. Rank heads by Davidson et al.'s causal score averaged over tasks, drop the generic heads among the top 30, and take the top $N$. It is the best set at every size; 20 heads already carry facts and morphology, and it keeps improving up to 80. On 30 other wordings of each task it stays the best set, with a smaller lead (Sec. 5.5).
3. **Without patching, per task: decoding, low BOS and attention to the instruction and its markers**, DBMN at 60 heads or DBTMS at 80 to 100 (Sec. 5.6). Chosen from J-head readings and attention alone, these are the best per-task sets of any size, above the per-task causal ranking. They were picked among 79 combinations on the test items, so a fresh test is still needed; of the sets fixed beforehand, DBT or BT at 40 heads come closest. On other wordings, DBT (the one run there) holds up best of the per-task rules.
4. **Choose on many wordings.** Sets chosen on all 35 templates beat those chosen on Yang's five, on the five themselves as well; only the causal ranking with its generic heads does not gain.
5. **Without patching, shared: DBMN or DBTMS at 80 to 120 heads**, level with Shared Causal − generic at its best (Sec. 5.6; picked on the test items). Of the sets fixed beforehand, DBT or BT at 60 heads, about level with the shared causal ranking with its generic heads at that size.
6. **Davidson et al.'s summed vector only without its generic heads**, and not beyond 20 heads; the same heads patched one by one do better from 40 heads on.
7. **By task type**: translations need per-task attention filters, or a shared causal set of at least 40 heads; morphology is carried best by causal rankings; facts by almost any rule from 20 heads; sequences by no set found here.
8. **Not**: Yang et al.'s hand-written lists, D or B alone, and DB with more than 40 heads per task.

# 7 Limitations

One model and 14 simple tasks. Every run is single and greedy, without intervals; rerunning a set with differently made-up batches moves its mean by up to 0.004 (Sec. 4), so differences of a point or two between sets are not established. Only $\alpha$ 1 and 2 are compared, which may underrate sets that peak higher. The generic-head rule was changed during the design. The rule fixed before the per-half runs (heads above the largest drop in the sorted generic score) became unstable once the score was computed on all five templates, where its two largest drops were nearly equal. We replaced it with the two-group split, choosing between the rules on the generic scores alone, not on accuracy. But the split marks the same heads as earlier per-half runs whose accuracies we had seen. Ranked combination sets at large $N$ hold heads that rank poorly on one filter. T has no task word in three templates (antonym, present–past), so on those tasks it rests on fewer prompts. The word lists keep some word pieces and Japanese fragments. Yang terms has fewer than $N$ heads on seven tasks at 60 and 80 heads. Items are held out, but wordings only in part: the sets chosen on Yang's five templates are tested on 30 templates they were not chosen on, whereas the sets chosen on all 35 are tested on the same 35, so part of their gain may be a closer fit to these templates. The 30 new templates per task were checked by script only, not by hand; 20 of them have no correctly answered prompt on one half, and on the six small tasks some means rest on fewer than 30 prompts (Table 12). The run on other wordings covers three rules at two sizes. The capped comparison runs check 1 only. The filters on other parts of the prompt (Sec. 5.6) were chosen among 79 combinations on the held-out accuracy they are reported on, and their categories are assigned by script; their lead is therefore not yet tested on fresh items.

### Conditions still to run

Wordings held out: sets chosen on some of the 35 templates and tested on the others. The other rules (BT, T, Davidson FV) and the other sizes on all 35 templates. The best combinations of Sec. 5.6 on fresh items (a new split, or chosen on one fold's accuracy and tested on the other) and on other wordings.

**Code and data.** The capped comparison: word lists `scripts/e13_lists.py`, attention pass `scripts/e13_attention.py`, sets and generic heads `scripts/e13_sets.py`, runs `scripts/e13_patch.sh`, which also runs Davidson et al.'s summed vector with `scripts/e13_fv_residual.py`, and tables and figures `scripts/e13_report.py`. Results in `results/e13_capped`: per fold the word lists, sets, generic heads, accuracies and every patched reply; `summary.csv` and `per_task.csv` at the cross-fitted $\alpha$. The causal scores are those of the first held-out runs (`results/heldout/fold{1,2}/fv`). Other wordings: templates `materials/simple/e4b_wordings.json`, unpatched accuracy `scripts/e14_outputs.py`, word lists `scripts/e14_lists.py`, attention `scripts/e13_attention.py --e14`, causal scores `scripts/davidson_fv.py --instructions all`, sets `scripts/e14_sets.py`, runs `scripts/e14_run.sh`, tables `scripts/e14_report.py`; results in `results/e14_wordings` (report `paper/e14-all-wordings`). Other parts of the prompt: token categories and attention `scripts/e20_bai_attention.py` (stopword list `materials/stopwords/nltk_english.txt`), sets `scripts/e22_sets.py` and `scripts/e23_sets.py`, runs `scripts/e22_patch.sh` and `scripts/e23_patch.sh`, tables `scripts/e22_report.py` and `scripts/e23_report.py`, figures `scripts/e23_figures.py`; results in `results/e22_all_combos` and `results/e23_best_sizes`. Table 2: `scripts/synthesis_report.py`, from the saved result files.

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# A Per-task results

| | | One set per task | | | | | | | One set for all tasks | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Task | Unp. | DBT | DB | T | DBT −gen. | Causal | Causal −gen. | Yang | DBT | DBT −gen. | Causal | Causal −gen. | FV sum | FV sum −gen. | Yang |
| *Word relations* | | | | | | | | | | | | | | | |
| antonym | 0.00 | 0.41 | **0.59** | 0.38 | 0.50 | 0.23 | 0.44 | 0.01 | 0.24 | 0.19 | 0.26 | 0.45 | 0.31 | 0.32 | 0.00 |
| synonym | 0.03 | 0.30 | 0.31 | 0.22 | 0.23 | 0.28 | **0.51** | 0.16 | 0.12 | 0.06 | 0.22 | 0.25 | 0.04 | 0.20 | 0.01 |
| *Morphology* | | | | | | | | | | | | | | | |
| singular–plural | 0.00 | 0.79 | 0.85 | 0.53 | 0.71 | 0.62 | 0.80 | 0.01 | 0.79 | 0.74 | 0.77 | **0.87** | 0.68 | 0.83 | 0.03 |
| present–past | 0.00 | 0.50 | 0.55 | 0.00 | 0.42 | 0.68 | **0.76** | 0.00 | 0.27 | 0.41 | 0.69 | 0.76 | 0.04 | 0.41 | 0.00 |
| *Facts* | | | | | | | | | | | | | | | |
| country–capital | 0.00 | 0.97 | 0.85 | 0.97 | 0.97 | 0.92 | 0.97 | 0.71 | 0.97 | 0.93 | 0.95 | **0.98** | 0.64 | 0.96 | 0.63 |
| country–currency | 0.00 | 0.47 | 0.45 | 0.45 | 0.41 | 0.13 | 0.10 | 0.13 | 0.64 | 0.62 | **0.67** | 0.60 | 0.51 | 0.61 | 0.07 |
| landmark–country | 0.01 | 0.95 | 0.30 | 0.93 | 0.94 | 0.96 | 0.95 | 0.68 | 0.96 | 0.94 | 0.86 | 0.93 | 0.93 | **0.96** | 0.67 |
| park–country | 0.00 | 0.78 | 0.51 | 0.55 | 0.69 | **0.89** | 0.84 | 0.47 | 0.88 | 0.79 | 0.68 | 0.82 | 0.86 | 0.89 | 0.49 |
| product–company | 0.07 | 0.84 | 0.70 | 0.81 | 0.76 | 0.77 | **0.90** | 0.37 | 0.90 | 0.85 | 0.80 | 0.88 | 0.83 | 0.87 | 0.63 |
| *Sequences* | | | | | | | | | | | | | | | |
| next-item | 0.00 | 0.10 | 0.01 | 0.05 | 0.06 | 0.15 | 0.04 | 0.02 | 0.14 | 0.10 | 0.10 | 0.10 | **0.22** | 0.04 | 0.02 |
| prev-item | 0.00 | **0.28** | 0.01 | 0.07 | 0.27 | 0.03 | 0.14 | 0.08 | 0.18 | 0.16 | 0.14 | 0.23 | 0.07 | 0.03 | 0.06 |
| *Translations* | | | | | | | | | | | | | | | |
| English–French | 0.00 | **0.46** | 0.39 | 0.45 | 0.30 | 0.06 | 0.03 | 0.04 | 0.12 | 0.06 | 0.35 | 0.23 | 0.21 | 0.16 | 0.00 |
| English–German | 0.00 | 0.34 | 0.31 | 0.29 | 0.23 | 0.00 | 0.00 | 0.01 | 0.06 | 0.04 | **0.37** | 0.19 | 0.08 | 0.14 | 0.00 |
| English–Spanish | 0.00 | 0.55 | 0.51 | 0.54 | 0.43 | 0.01 | 0.01 | 0.01 | 0.24 | 0.10 | **0.59** | 0.34 | 0.27 | 0.37 | 0.00 |
| Mean | 0.01 | 0.55 | 0.45 | 0.45 | 0.49 | 0.41 | 0.46 | 0.19 | 0.46 | 0.43 | 0.53 | 0.55 | 0.41 | 0.49 | 0.19 |

**Table 9:** Check 1 per task at 40 heads, held-out items, cross-fitted $\alpha \in \{1, 2\}$, mean of the two folds. Unp.: unpatched accuracy. Bold: the row's highest value.

| | One set per task | | | | | | One set for all 14 tasks | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Heads (N) | 20 | 40 | 60 | 80 | 100 | 120 | 20 | 40 | 60 | 80 | 100 | 120 |
| *Decoding and attention filters* | | | | | | | | | | | | |
| D | 0.03 | 0.02 | 0.02 | 0.02 | 0.02 | 0.02 | 0.01 | 0.01 | 0.01 | 0.02 | 0.02 | 0.02 |
| B | 0.00 | 0.00 | 0.01 | 0.01 | 0.00 | 0.00 | 0.01 | 0.01 | 0.02 | 0.02 | 0.02 | 0.01 |
| T | 0.03 | 0.03 | 0.03 | 0.04 | 0.02 | 0.03 | 0.02 | 0.03 | 0.02 | 0.02 | 0.01 | 0.01 |
| DB | 0.02 | 0.01 | 0.01 | 0.01 | 0.01 | 0.01 | 0.02 | 0.01 | 0.01 | 0.01 | 0.01 | 0.01 |
| DT | 0.04 | 0.03 | 0.02 | 0.01 | 0.01 | 0.01 | 0.02 | 0.01 | 0.01 | 0.01 | 0.01 | 0.01 |
| BT | 0.04 | 0.01 | 0.02 | 0.02 | 0.01 | 0.02 | 0.05 | 0.02 | 0.02 | 0.01 | 0.02 | 0.01 |
| DBT | 0.02 | 0.02 | 0.01 | 0.01 | 0.01 | 0.02 | 0.02 | 0.01 | 0.01 | 0.01 | 0.01 | 0.01 |
| DBT − generic | 0.02 | 0.01 | 0.01 | 0.01 | 0.01 | 0.01 | 0.02 | 0.01 | 0.01 | 0.01 | 0.01 | 0.01 |
| *Davidson et al. (causal score)* | | | | | | | | | | | | |
| Causal | 0.03 | 0.03 | 0.03 | 0.03 | 0.03 | 0.04 | 0.03 | 0.04 | 0.02 | 0.02 | 0.02 | 0.02 |
| Causal − generic | 0.04 | 0.02 | 0.02 | 0.02 | 0.02 | 0.02 | 0.03 | 0.02 | 0.01 | 0.02 | 0.01 | 0.01 |
| Davidson FV (summed) | – | – | – | – | – | – | 0.03 | 0.04 | 0.04 | 0.03 | 0.03 | 0.03 |
| Davidson FV − generic (summed) | – | – | – | – | – | – | 0.03 | 0.03 | 0.04 | 0.05 | 0.03 | 0.02 |
| *Baseline* | | | | | | | | | | | | |
| Yang terms | 0.03 | 0.03 | 0.02 | 0.01 | – | – | 0.04 | 0.05 | 0.02 | 0.01 | 0.01 | 0.01 |
| *Control* | | | | | | | | | | | | |
| Random (DBT layers) | – | – | – | – | – | – | – | – | – | – | – | – |

**Table 10:** Control: the sets of Table 5 patched with the control task's mean (Sec. 2), check 1, mean over the 14 tasks, at the task mean's cross-fitted $\alpha$. Random sets are not run with the control mean.

# B Task-word lists

The 50 generic words removed from every list, from WikiText-103 (Sec. 4), by share: [Arabic letter], [Devanagari letter], pří, which, mph, http, his, what, https, whilst, pretty, colour, they, their, even, maybe, amongst,

programme, only, himself, she, bindingutil, really, again, her, then, also, herself, itself, organisation, sfml, however, behaviors, during, 1, [Thai repetition mark], our, being, ones, [Thai 'word'], later, there, him, [Korean 'date'], 页面存档备份, programmes, who, why, local, etc. Five of them are not in a Latin, Chinese, Japanese or Cyrillic script and are described in brackets.

**Table 11:** Task-word lists (fold 1, by share) and the words of each list that occur in the instructions, which T measures attention to. Last column: fold 2's list against fold 1's.

| Task | T words | 30-word list | Fold 2 |
|---|---|---|---|
| antonym | opposite, question, word | opposite, word, instead, because, opposing, words, therefore, definition, reverse, synonyms, whats, name, oppos, meaning, american, ant, answer, synonym, question, likely, adjective, thats, probably, 2, similar, vocabulary, above, either, same, how | +english, definitions, 3; −above, same, how |
| synonym | synonym, word | synonyms, word, words, 2, similar, synonym, definition, name, whats, english, syn, 3, same, afterward, possibly, instead, called, because, other, american, similarly, therefore, 4, somebody, answer, stuff, how, 5, others, named | +definitions; −named |
| singular–plural | plural | because, single, name, noun, multiple, plural, lingu, either, linguistic, therefore, names, named, naming, similar, english, called, whats, individual, 2, how, vocabulary, when, same, 3, nouns, identical, sisters, pronunciation, question, very | +answer; −very |
| present–past | english, verb, word | english, verb, lingu, grammar, name, because, linguistic, question, verbs, gramm, called, word, how, britain, whats, british, 2, language, named, instead, took, sentence, when, either, very, tom, transitioning, did, noun, therefore | +england; −did |
| country–capital | answer, capital, city, country | city, cities, country, town, countries, answer, international, geography, downtown, capital, canada, american, answers, italy, london, diplomatic, africa, location, città, germany, towns, political, question, named, therefore, usa, very, australia, because, england | +france, wikipedia; −germany, usa |
| country–currency | answer, country, currency | currency, currencies, country, monetary, money, countries, dollar, international, financial, american, answer, foreign, germany, british, canada, wikipedia, italy, instead, dollars, france, called, nationality, australia, ンツ, england, tom, nation, economic, netherlands, かし | +australian, 2, named, very; −england, nation, netherlands, かし |
| landmark–country | country, located, where | country, countries, england, canada, germany, france, location, australia, city, international, geography, italy, located, geographical, locations, usa, county, geographic, cities, netherlands, ireland, american, regional, town, named, wikipedia, british, where, nationality, japan | +tourism, america; −british, nationality |
| park–country | country, located, nation, park | country, countries, england, canada, location, australia, park, located, france, national, tourism, germany, geography, unesco, parks, international, italy, city, nation, tourist, geographical, county, wildlife, locations, british, named, nationality, tourists, american, geographic | +usa; −tourists |
| product–company | company, product | company, companies, manufacturer, product, microsoft, manufacturers, products, because, named, name, owned, wikipedia, software, apple, ownership, respectively, answer, manufacturing, mitsubishi, ibm, how, ーダ, question, very, probably, windows, 2, japanese, nintendo, production | +computer; −japanese |
| next-item | after, item, next, sequence | next, january, july, september, november, october, february, sequence, because, question, items, december, june, after, item, april, 2, calendar, answer, christmas, seven, when, 3, math, eight, monday, six, nine, each, named | +sunday, friday, tuesday, wednesday, saturday, thursday, 4, answers; −february, june, april, seven, eight, six, each, named |

| Task | T words | 30-word list | Fold 2 |
|---|---|---|---|
| prev-item | before, item, previous, return, sequence | previous, preceding, prior, answer, question, item, next, because, items, sequence, january, previously, earlier, september, answers, how, november, october, before, preced, questions, july, 2, return, february, seven, when, therefore, whats, december | +monday, sunday, 3, christmas; −february, seven, therefore, whats |
| English–French | english, french, translation; fold 2: english, french, into, translation | french, france, english, paris, 法国, translation, italian, european, american, француз, translations, language, linguistic, japanese, german, françois, called, bilingual, therefore, languages, how, named, name, quebec, translated, foreign, portuguese, lingu, franc, 2 | +into, behaviour; −foreign, 2 |
| English–German | english, german, translation | german, germany, germans, english, european, berlin, french, american, translation, linguistic, language, deutschland, called, therefore, italian, von, japanese, munich, czech, translations, languages, lingu, die, bav, austria, bilingual, named, name, auf, austrian | +british; −austrian |
| English–Spanish | english, spanish, translation | spanish, english, mexican, hispanic, translation, bilingual, spain, language, mexico, translations, portuguese, linguistic, french, ilingual, called, languages, american, name, maría, madrid, santiago, therefore, japanese, because, lingu, italian, méxico, named, translated, synonyms | none |

# C Other wordings

| | E0 accuracy (flexible), mean of the group's templates | | | | | | | Templates | |
|---|---|---|---|---|---|---|---|---|---|
| Task | Yang's 5 | cloze, named | cloze, not named | Q–A, named | Q–A, not named | label, named | label, not named | $< 30$ | 0 |
| antonym | 0.68 | 0.37 | 0.52 | 0.67 | 0.68 | 0.67 | 0.67 | 1 | 0 |
| synonym | 0.45 | 0.30 | 0.30 | 0.48 | 0.49 | 0.20 | 0.23 | 2 | 1 |
| singular–plural | 0.98 | 0.58 | 0.23 | 0.99 | 0.96 | 0.99 | 0.94 | 5 | 3 |
| present–past | 0.96 | 0.39 | 0.39 | 0.98 | 0.93 | 0.99 | 0.65 | 6 | 4 |
| country–capital | 0.97 | 0.70 | 0.77 | 0.97 | 0.94 | 0.97 | 0.93 | 2 | 1 |
| country–currency | 0.72 | 0.45 | 0.66 | 0.68 | 0.45 | 0.66 | 0.10 | 9 | 5 |
| landmark–country | 0.85 | 0.55 | 0.66 | 0.83 | 0.82 | 0.70 | 0.50 | 1 | 1 |
| park–country | 0.79 | 0.57 | 0.32 | 0.89 | 0.84 | 0.89 | 0.61 | 1 | 0 |
| product–company | 0.73 | 0.58 | 0.57 | 0.73 | 0.68 | 0.74 | 0.69 | 0 | 0 |
| next-item | 0.42 | 0.44 | 0.32 | 0.39 | 0.20 | 0.50 | 0.64 | 12 | 1 |
| prev-item | 0.53 | 0.29 | 0.32 | 0.54 | 0.35 | 0.46 | 0.42 | 11 | 0 |
| English–French | 0.73 | 0.53 | 0.22 | 0.78 | 0.64 | 0.73 | 0.71 | 1 | 0 |
| English–German | 0.67 | 0.49 | 0.22 | 0.68 | 0.46 | 0.73 | 0.40 | 5 | 3 |
| English–Spanish | 0.79 | 0.59 | 0.46 | 0.82 | 0.74 | 0.71 | 0.82 | 1 | 1 |
| *Mean* | 0.73 | 0.49 | 0.43 | 0.75 | 0.65 | 0.71 | 0.59 | 57 | 20 |

**Table 12:** Unpatched accuracy of the 35 templates (flexible score), mean of each group's five templates, on the first 500 items of each half. Templates: $< 30$, with fewer than 30 correctly answered prompts on at least one half; 0, with none on at least one half, which give no mean there.

| | | Means from | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Set, heads chosen on | All 35 | Yang's 5 | cloze, named | cloze, not named | Q–A, named | Q–A, not named | label, named | label, not named |
| *One set per task* | | | | | | | | |
| DBT-40, Yang | 0.50 | 0.55 | 0.44 | 0.34 | 0.57 | 0.49 | 0.58 | 0.46 |
| DBT-40, all | 0.57 | 0.63 | 0.53 | 0.42 | 0.63 | 0.54 | 0.66 | 0.55 |
| Causal-40, Yang | 0.37 | 0.43 | 0.28 | 0.22 | 0.48 | 0.42 | 0.41 | 0.30 |
| Causal-40, all | 0.40 | 0.46 | 0.30 | 0.24 | 0.53 | 0.46 | 0.44 | 0.32 |
| Causal − generic-40, Yang | 0.40 | 0.47 | 0.36 | 0.28 | 0.49 | 0.43 | 0.43 | 0.32 |
| Causal − generic-40, all | 0.49 | 0.54 | 0.44 | 0.38 | 0.58 | 0.52 | 0.51 | 0.41 |
| DBT-80, Yang | 0.49 | 0.55 | 0.43 | 0.30 | 0.57 | 0.47 | 0.58 | 0.47 |
| DBT-80, all | 0.57 | 0.64 | 0.53 | 0.41 | 0.65 | 0.54 | 0.64 | 0.51 |
| Causal-80, Yang | 0.44 | 0.53 | 0.33 | 0.26 | 0.59 | 0.47 | 0.49 | 0.36 |
| Causal-80, all | 0.48 | 0.56 | 0.38 | 0.33 | 0.64 | 0.55 | 0.48 | 0.37 |
| Causal − generic-80, Yang | 0.45 | 0.52 | 0.40 | 0.34 | 0.56 | 0.43 | 0.49 | 0.38 |
| Causal − generic-80, all | 0.52 | 0.58 | 0.48 | 0.42 | 0.63 | 0.54 | 0.52 | 0.42 |
| DBT-100, Yang | 0.43 | 0.48 | 0.37 | 0.26 | 0.54 | 0.44 | 0.51 | 0.38 |
| DBT-100, all | 0.57 | 0.63 | 0.51 | 0.39 | 0.66 | 0.56 | 0.65 | 0.52 |
| Causal-100, Yang | 0.47 | 0.56 | 0.35 | 0.28 | 0.62 | 0.50 | 0.51 | 0.38 |
| Causal-100, all | 0.52 | 0.60 | 0.41 | 0.37 | 0.67 | 0.59 | 0.53 | 0.39 |
| Causal − generic-100, Yang | 0.44 | 0.52 | 0.39 | 0.31 | 0.55 | 0.45 | 0.47 | 0.35 |
| Causal − generic-100, all | 0.54 | 0.60 | 0.49 | 0.42 | 0.66 | 0.57 | 0.55 | 0.42 |
| DBT-120, Yang | 0.37 | 0.41 | 0.29 | 0.21 | 0.48 | 0.38 | 0.44 | 0.32 |
| DBT-120, all | 0.55 | 0.62 | 0.47 | 0.35 | 0.66 | 0.58 | 0.61 | 0.50 |
| Causal-120, Yang | 0.51 | 0.61 | 0.40 | 0.33 | 0.65 | 0.56 | 0.55 | 0.42 |
| Causal-120, all | 0.53 | 0.62 | 0.41 | 0.38 | 0.68 | 0.61 | 0.55 | 0.42 |
| Causal − generic-120, Yang | 0.48 | 0.56 | 0.42 | 0.34 | 0.58 | 0.48 | 0.52 | 0.40 |
| Causal − generic-120, all | 0.55 | 0.62 | 0.47 | 0.42 | 0.68 | 0.60 | 0.57 | 0.44 |
| *One set for all 14 tasks* | | | | | | | | |
| DBT-40, Yang | 0.43 | 0.46 | 0.40 | 0.33 | 0.46 | 0.43 | 0.49 | 0.40 |
| DBT-40, all | 0.55 | 0.59 | 0.52 | 0.45 | 0.59 | 0.54 | 0.60 | 0.53 |
| Causal-40, Yang | 0.46 | 0.53 | 0.29 | 0.23 | 0.61 | 0.52 | 0.54 | 0.39 |
| Causal-40, all | 0.45 | 0.54 | 0.27 | 0.22 | 0.64 | 0.53 | 0.50 | 0.34 |
| Causal − generic-40, Yang | 0.47 | 0.55 | 0.40 | 0.34 | 0.59 | 0.49 | 0.51 | 0.38 |
| Causal − generic-40, all | 0.57 | 0.64 | 0.49 | 0.43 | 0.69 | 0.59 | 0.61 | 0.52 |
| DBT-80, Yang | 0.49 | 0.55 | 0.43 | 0.34 | 0.56 | 0.48 | 0.58 | 0.47 |
| DBT-80, all | 0.57 | 0.63 | 0.51 | 0.41 | 0.66 | 0.54 | 0.65 | 0.56 |
| Causal-80, Yang | 0.56 | 0.65 | 0.40 | 0.34 | 0.71 | 0.61 | 0.63 | 0.49 |
| Causal-80, all | 0.54 | 0.63 | 0.38 | 0.33 | 0.71 | 0.58 | 0.61 | 0.47 |
| Causal − generic-80, Yang | 0.61 | 0.69 | 0.54 | 0.48 | 0.73 | 0.62 | 0.65 | 0.54 |
| Causal − generic-80, all | 0.63 | 0.70 | 0.57 | 0.51 | 0.74 | 0.62 | 0.68 | 0.58 |
| DBT-100, Yang | 0.46 | 0.51 | 0.38 | 0.29 | 0.55 | 0.45 | 0.54 | 0.44 |
| DBT-100, all | 0.56 | 0.64 | 0.50 | 0.38 | 0.66 | 0.54 | 0.65 | 0.53 |
| Causal-100, Yang | 0.59 | 0.69 | 0.45 | 0.40 | 0.73 | 0.62 | 0.65 | 0.51 |
| Causal-100, all | 0.57 | 0.67 | 0.44 | 0.39 | 0.73 | 0.63 | 0.62 | 0.47 |
| Causal − generic-100, Yang | 0.62 | 0.71 | 0.57 | 0.52 | 0.73 | 0.62 | 0.62 | 0.52 |
| Causal − generic-100, all | 0.65 | 0.72 | 0.63 | 0.55 | 0.74 | 0.64 | 0.67 | 0.57 |
| DBT-120, Yang | 0.46 | 0.52 | 0.37 | 0.26 | 0.58 | 0.48 | 0.54 | 0.44 |
| DBT-120, all | 0.59 | 0.67 | 0.51 | 0.38 | 0.71 | 0.60 | 0.67 | 0.55 |
| Causal-120, Yang | 0.60 | 0.71 | 0.47 | 0.42 | 0.76 | 0.66 | 0.65 | 0.51 |
| Causal-120, all | 0.60 | 0.70 | 0.47 | 0.44 | 0.74 | 0.65 | 0.64 | 0.49 |
| Causal − generic-120, Yang | 0.58 | 0.69 | 0.53 | 0.47 | 0.70 | 0.59 | 0.57 | 0.48 |
| Causal − generic-120, all | 0.61 | 0.69 | 0.59 | 0.50 | 0.71 | 0.61 | 0.63 | 0.53 |

**Table 13:** Check 1 by the group of the template whose mean is patched, mean over the 14 tasks, held-out items, mean of the two folds, at the $\alpha$ cross-fitted on all 35 means. Yang, all: heads chosen on Yang's five or on all 35 templates.

| | | One set per task | | | | | | One set for all tasks | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | DBT | | Causal | | Causal −g. | | DBT | | Causal | | Causal −g. | |
| Task | Unp. | Y5 | A35 | Y5 | A35 | Y5 | A35 | Y5 | A35 | Y5 | A35 | Y5 | A35 |
| *40 heads* | | | | | | | | | | | | | |
| antonym | 0.00 | 0.37 | 0.49 | 0.30 | 0.34 | 0.36 | 0.60 | 0.25 | 0.48 | 0.28 | 0.25 | 0.39 | 0.52 |
| synonym | 0.03 | 0.28 | 0.43 | 0.24 | 0.30 | 0.42 | 0.47 | 0.16 | 0.34 | 0.22 | 0.20 | 0.25 | 0.34 |
| singular–plural | 0.00 | 0.76 | 0.83 | 0.71 | 0.57 | 0.73 | 0.80 | 0.78 | 0.85 | 0.73 | 0.74 | 0.79 | 0.88 |
| present–past | 0.00 | 0.40 | 0.43 | 0.45 | 0.41 | 0.49 | 0.67 | 0.25 | 0.66 | 0.45 | 0.54 | 0.52 | 0.66 |
| country–capital | 0.00 | 0.95 | 0.96 | 0.80 | 0.86 | 0.93 | 0.94 | 0.89 | 0.96 | 0.88 | 0.88 | 0.94 | 0.95 |
| country–currency | 0.00 | 0.36 | 0.54 | 0.08 | 0.10 | 0.06 | 0.17 | 0.57 | 0.66 | 0.47 | 0.42 | 0.43 | 0.56 |
| landmark–country | 0.01 | 0.88 | 0.89 | 0.91 | 0.86 | 0.88 | 0.91 | 0.90 | 0.90 | 0.78 | 0.73 | 0.86 | 0.88 |
| park–country | 0.00 | 0.73 | 0.79 | 0.81 | 0.83 | 0.74 | 0.81 | 0.79 | 0.80 | 0.63 | 0.58 | 0.70 | 0.77 |
| product–company | 0.07 | 0.76 | 0.76 | 0.64 | 0.67 | 0.85 | 0.89 | 0.82 | 0.86 | 0.67 | 0.60 | 0.82 | 0.84 |
| next-item | 0.00 | 0.11 | 0.16 | 0.10 | 0.01 | 0.05 | 0.01 | 0.09 | 0.17 | 0.09 | 0.09 | 0.08 | 0.15 |
| prev-item | 0.00 | 0.14 | 0.16 | 0.01 | 0.03 | 0.05 | 0.06 | 0.09 | 0.17 | 0.06 | 0.07 | 0.12 | 0.18 |
| English–French | 0.00 | 0.39 | 0.45 | 0.05 | 0.15 | 0.02 | 0.13 | 0.12 | 0.25 | 0.33 | 0.34 | 0.19 | 0.38 |
| English–German | 0.00 | 0.29 | 0.43 | 0.02 | 0.09 | 0.02 | 0.07 | 0.07 | 0.16 | 0.28 | 0.30 | 0.14 | 0.31 |
| English–Spanish | 0.00 | 0.52 | 0.65 | 0.04 | 0.36 | 0.02 | 0.29 | 0.25 | 0.46 | 0.51 | 0.49 | 0.35 | 0.59 |
| *Mean* | 0.01 | 0.50 | 0.57 | 0.37 | 0.40 | 0.40 | 0.49 | 0.43 | 0.55 | 0.46 | 0.45 | 0.47 | 0.57 |
| *80 heads* | | | | | | | | | | | | | |
| antonym | 0.00 | 0.57 | 0.69 | 0.38 | 0.42 | 0.46 | 0.65 | 0.33 | 0.51 | 0.47 | 0.48 | 0.60 | 0.64 |
| synonym | 0.03 | 0.20 | 0.38 | 0.42 | 0.48 | 0.41 | 0.55 | 0.21 | 0.27 | 0.36 | 0.34 | 0.39 | 0.39 |
| singular–plural | 0.00 | 0.56 | 0.81 | 0.54 | 0.74 | 0.68 | 0.81 | 0.72 | 0.78 | 0.82 | 0.84 | 0.90 | 0.92 |
| present–past | 0.00 | 0.58 | 0.48 | 0.48 | 0.42 | 0.40 | 0.44 | 0.36 | 0.65 | 0.64 | 0.55 | 0.74 | 0.70 |
| country–capital | 0.00 | 0.73 | 0.94 | 0.89 | 0.91 | 0.94 | 0.94 | 0.88 | 0.94 | 0.91 | 0.90 | 0.93 | 0.93 |
| country–currency | 0.00 | 0.55 | 0.61 | 0.24 | 0.29 | 0.24 | 0.33 | 0.58 | 0.66 | 0.49 | 0.47 | 0.52 | 0.57 |
| landmark–country | 0.01 | 0.84 | 0.85 | 0.86 | 0.86 | 0.90 | 0.91 | 0.87 | 0.85 | 0.80 | 0.81 | 0.88 | 0.90 |
| park–country | 0.00 | 0.67 | 0.81 | 0.83 | 0.79 | 0.78 | 0.76 | 0.75 | 0.73 | 0.68 | 0.68 | 0.77 | 0.81 |
| product–company | 0.07 | 0.74 | 0.72 | 0.76 | 0.77 | 0.89 | 0.90 | 0.77 | 0.76 | 0.72 | 0.69 | 0.83 | 0.83 |
| next-item | 0.00 | 0.09 | 0.14 | 0.31 | 0.01 | 0.16 | 0.02 | 0.20 | 0.17 | 0.18 | 0.14 | 0.26 | 0.23 |
| prev-item | 0.00 | 0.09 | 0.09 | 0.07 | 0.10 | 0.10 | 0.13 | 0.23 | 0.14 | 0.15 | 0.12 | 0.29 | 0.23 |
| English–French | 0.00 | 0.40 | 0.51 | 0.16 | 0.26 | 0.15 | 0.24 | 0.32 | 0.50 | 0.49 | 0.48 | 0.48 | 0.54 |
| English–German | 0.00 | 0.33 | 0.30 | 0.21 | 0.20 | 0.20 | 0.19 | 0.24 | 0.40 | 0.41 | 0.43 | 0.37 | 0.45 |
| English–Spanish | 0.00 | 0.51 | 0.61 | 0.03 | 0.46 | 0.01 | 0.40 | 0.44 | 0.64 | 0.64 | 0.63 | 0.65 | 0.72 |
| *Mean* | 0.01 | 0.49 | 0.57 | 0.44 | 0.48 | 0.45 | 0.52 | 0.49 | 0.57 | 0.56 | 0.54 | 0.61 | 0.63 |

**Table 14:** Check 1 per task with the means of all 35 templates, held-out items, cross-fitted $\alpha \in \{1, 2\}$, mean of the two folds. Y5, A35: heads chosen on Yang's five or on all 35 templates. Causal −g.: Causal − generic. Unp.: unpatched accuracy.

| | | One set per task | | | | | | One set for all tasks | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | DBT | | Causal | | Causal −g. | | DBT | | Causal | | Causal −g. | |
| Task | Unp. | Y5 | A35 | Y5 | A35 | Y5 | A35 | Y5 | A35 | Y5 | A35 | Y5 | A35 |
| *100 heads* | | | | | | | | | | | | | |
| antonym | 0.00 | 0.60 | 0.68 | 0.42 | 0.51 | 0.47 | 0.67 | 0.35 | 0.54 | 0.54 | 0.53 | 0.66 | 0.67 |
| synonym | 0.03 | 0.19 | 0.41 | 0.46 | 0.52 | 0.40 | 0.54 | 0.14 | 0.31 | 0.41 | 0.33 | 0.38 | 0.37 |
| singular–plural | 0.00 | 0.56 | 0.83 | 0.59 | 0.79 | 0.54 | 0.77 | 0.69 | 0.79 | 0.87 | 0.85 | 0.90 | 0.92 |
| present–past | 0.00 | 0.47 | 0.55 | 0.60 | 0.69 | 0.54 | 0.67 | 0.29 | 0.57 | 0.62 | 0.59 | 0.60 | 0.69 |
| country–capital | 0.00 | 0.65 | 0.93 | 0.91 | 0.90 | 0.94 | 0.93 | 0.85 | 0.92 | 0.92 | 0.92 | 0.93 | 0.93 |
| country–currency | 0.00 | 0.53 | 0.59 | 0.25 | 0.28 | 0.21 | 0.36 | 0.55 | 0.64 | 0.52 | 0.53 | 0.54 | 0.61 |
| landmark–country | 0.01 | 0.68 | 0.84 | 0.80 | 0.88 | 0.88 | 0.89 | 0.82 | 0.84 | 0.84 | 0.83 | 0.87 | 0.89 |
| park–country | 0.00 | 0.63 | 0.75 | 0.82 | 0.78 | 0.79 | 0.76 | 0.70 | 0.73 | 0.74 | 0.73 | 0.76 | 0.81 |
| product–company | 0.07 | 0.71 | 0.67 | 0.79 | 0.79 | 0.88 | 0.89 | 0.72 | 0.76 | 0.74 | 0.71 | 0.82 | 0.82 |
| next-item | 0.00 | 0.10 | 0.18 | 0.35 | 0.02 | 0.21 | 0.03 | 0.20 | 0.21 | 0.21 | 0.19 | 0.29 | 0.31 |
| prev-item | 0.00 | 0.02 | 0.09 | 0.08 | 0.09 | 0.13 | 0.16 | 0.18 | 0.18 | 0.18 | 0.17 | 0.32 | 0.26 |
| English–French | 0.00 | 0.32 | 0.43 | 0.17 | 0.23 | 0.16 | 0.23 | 0.25 | 0.47 | 0.52 | 0.54 | 0.51 | 0.58 |
| English–German | 0.00 | 0.19 | 0.31 | 0.24 | 0.28 | 0.01 | 0.22 | 0.19 | 0.37 | 0.45 | 0.47 | 0.40 | 0.49 |
| English–Spanish | 0.00 | 0.41 | 0.67 | 0.06 | 0.45 | 0.03 | 0.40 | 0.45 | 0.58 | 0.65 | 0.64 | 0.67 | 0.73 |
| *Mean* | 0.01 | 0.43 | 0.57 | 0.47 | 0.52 | 0.44 | 0.54 | 0.46 | 0.56 | 0.59 | 0.57 | 0.62 | 0.65 |
| *120 heads* | | | | | | | | | | | | | |
| antonym | 0.00 | 0.57 | 0.66 | 0.45 | 0.55 | 0.51 | 0.68 | 0.33 | 0.55 | 0.56 | 0.56 | 0.63 | 0.64 |
| synonym | 0.03 | 0.15 | 0.40 | 0.47 | 0.55 | 0.40 | 0.54 | 0.16 | 0.33 | 0.43 | 0.37 | 0.26 | 0.33 |
| singular–plural | 0.00 | 0.46 | 0.84 | 0.75 | 0.81 | 0.59 | 0.78 | 0.70 | 0.84 | 0.90 | 0.89 | 0.88 | 0.87 |
| present–past | 0.00 | 0.29 | 0.61 | 0.69 | 0.76 | 0.62 | 0.75 | 0.34 | 0.66 | 0.70 | 0.65 | 0.40 | 0.52 |
| country–capital | 0.00 | 0.51 | 0.94 | 0.91 | 0.90 | 0.94 | 0.93 | 0.86 | 0.92 | 0.93 | 0.93 | 0.91 | 0.92 |
| country–currency | 0.00 | 0.49 | 0.49 | 0.28 | 0.33 | 0.26 | 0.41 | 0.56 | 0.61 | 0.53 | 0.54 | 0.55 | 0.62 |
| landmark–country | 0.01 | 0.49 | 0.57 | 0.80 | 0.89 | 0.88 | 0.88 | 0.81 | 0.87 | 0.85 | 0.83 | 0.85 | 0.83 |
| park–country | 0.00 | 0.54 | 0.71 | 0.83 | 0.74 | 0.80 | 0.70 | 0.64 | 0.75 | 0.75 | 0.75 | 0.74 | 0.79 |
| product–company | 0.07 | 0.66 | 0.69 | 0.83 | 0.80 | 0.88 | 0.88 | 0.66 | 0.78 | 0.75 | 0.72 | 0.81 | 0.78 |
| next-item | 0.00 | 0.12 | 0.18 | 0.42 | 0.02 | 0.24 | 0.03 | 0.18 | 0.22 | 0.22 | 0.24 | 0.30 | 0.32 |
| prev-item | 0.00 | 0.01 | 0.10 | 0.08 | 0.13 | 0.12 | 0.19 | 0.15 | 0.21 | 0.20 | 0.18 | 0.30 | 0.24 |
| English–French | 0.00 | 0.24 | 0.49 | 0.19 | 0.28 | 0.22 | 0.32 | 0.34 | 0.49 | 0.51 | 0.54 | 0.49 | 0.54 |
| English–German | 0.00 | 0.15 | 0.31 | 0.25 | 0.26 | 0.22 | 0.22 | 0.26 | 0.38 | 0.43 | 0.47 | 0.37 | 0.45 |
| English–Spanish | 0.00 | 0.46 | 0.69 | 0.19 | 0.44 | 0.03 | 0.38 | 0.52 | 0.67 | 0.68 | 0.70 | 0.64 | 0.71 |
| *Mean* | 0.01 | 0.37 | 0.55 | 0.51 | 0.53 | 0.48 | 0.55 | 0.46 | 0.59 | 0.60 | 0.60 | 0.58 | 0.61 |

**Table 15:** As Table 14, at 100 and 120 heads.

| | One set per task | | | One set for all 14 tasks | | |
|---|---|---|---|---|---|---|
| Heads (N) | DBT | Causal | Causal − generic | DBT | Causal | Causal − generic |
| 40 | 29.3 | 23.6 | 22.0 | 35.5 | 33.0 | 30.0 |
| 80 | 58.7 | 46.6 | 44.6 | 61.0 | 65.5 | 63.0 |
| 100 | 73.1 | 58.5 | 55.7 | 79.0 | 83.5 | 77.0 |
| 120 | 88.1 | 68.9 | 66.3 | 93.0 | 94.5 | 91.0 |

**Table 16:** Heads of each set chosen on all 35 templates that are also in the same set chosen on Yang's five, mean over tasks and folds.

Generic heads marked on all 35 templates, with those marked on Yang's five in brackets and the number of heads of Davidson et al.'s shared top 30 that stay: fold 1: L13H25, L13H31, L13H6, L14H29, L15H5, L16H29,

L28H10, L31H14 (on Yang's 5: L12H9, L13H31, L14H11, L14H29, L15H5, L15H7, L16H29, L28H10, L31H14; Davidson top 30 shared: 26); fold 2: L13H31, L13H6, L14H29, L15H5, L16H29, L29H23, L31H14 (on Yang's 5: L12H9, L13H31, L14H11, L14H29, L15H5, L15H7, L16H29, L31H14; Davidson top 30 shared: 25).

# D Davidson et al.'s FV on all items

These runs precede the held-out protocol: heads, means and generic heads were read on all items, so they are not held out. With the 30 heads chosen over the 14 tasks, the FV beats our best lexical-head set on 11 of 14 tasks, including tasks the lexical sets do not carry (singular–plural, antonym). Six of the 30 heads were marked generic, read on all items. Without these six (24 heads) the FV is at least as good on all 14 tasks, works best at $\alpha$ 2–3, and also carries the translations. Removing six random FV heads instead does not help.